

Task 2: Data Analysis Using Pandas

Name: Kandukuri Bhanu Sri

Candidate ID: BS/REG/122711

Domain: Python Developer Intern

Task Objective

Demonstrate data analysis skills using the Pandas library by loading a CSV dataset, cleaning missing data, applying filtering, grouping, and aggregation, and summarizing the insights.

Requirements Completed

- ✓ Loaded and inspected a CSV dataset
 - ✓ Cleaned missing or incorrect data
 - ✓ Applied filtering on the dataset
 - ✓ Performed grouping and aggregation using Pandas
 - ✓ Explained the insights in simple words
-

Dataset

Dataset Name: sample_data.csv

Description:

The dataset contains employee information such as Employee ID, Name, Department, Salary, and Age.

ID,Name,Department,Salary,Age

101,Alice,HR,50000,25

102,Bob,IT,60000,28

103,Charlie,IT,,30

104,David,Finance,55000,29

105,Eva,HR,52000,

106,Frank,Finance,58000,31

107,Grace,IT,65000,27

Python Code

```
import pandas as pd

# Load the CSV dataset
df = pd.read_csv("sample_data.csv")

# Display first five rows
print("Dataset:")
print(df.head())

# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Clean missing data
df["Salary"] = df["Salary"].fillna(df["Salary"].mean())
df["Age"] = df["Age"].fillna(df["Age"].mean())

print("\nCleaned Dataset:")
print(df)

# Filter employees with salary greater than 55000
high_salary = df[df["Salary"] > 55000]

print("\nEmployees with Salary > 55000:")
print(high_salary)

# Group by Department and calculate average salary
grouped = df.groupby("Department")["Salary"].mean()

print("\nAverage Salary by Department:")
```

```
print(grouped)
```

```
# Overall statistics
```

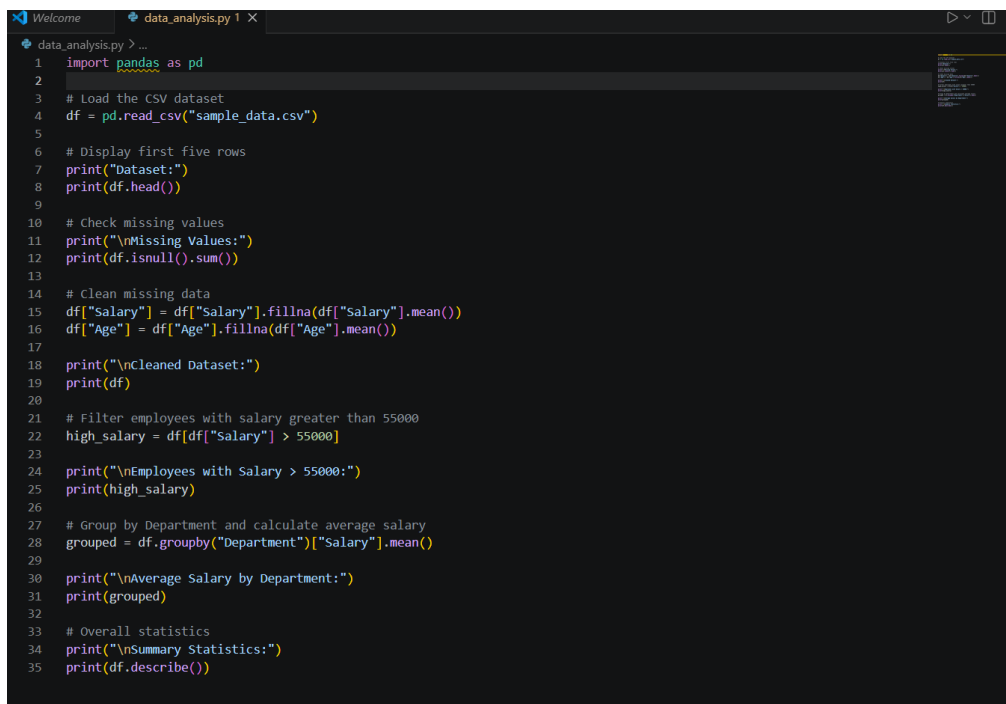
```
print("\nSummary Statistics:")
```

```
print(df.describe())
```

Screenshots

Screenshot 1

Python script in VS Code/IDLE

A screenshot of a Python script in a VS Code editor. The script is named 'data_analysis.py' and contains 35 lines of code. The code performs various data analysis tasks using pandas, including loading a CSV file, displaying the first five rows, checking for missing values, cleaning missing data, filtering employees with salary greater than 55000, grouping by department to calculate average salary, and printing overall statistics. The script is shown in a dark-themed editor with a file explorer on the right.

```
1 import pandas as pd
2
3 # Load the CSV dataset
4 df = pd.read_csv("sample_data.csv")
5
6 # Display first five rows
7 print("Dataset:")
8 print(df.head())
9
10 # Check missing values
11 print("\nMissing Values:")
12 print(df.isnull().sum())
13
14 # Clean missing data
15 df["Salary"] = df["Salary"].fillna(df["Salary"].mean())
16 df["Age"] = df["Age"].fillna(df["Age"].mean())
17
18 print("\nCleaned Dataset:")
19 print(df)
20
21 # Filter employees with salary greater than 55000
22 high_salary = df[df["Salary"] > 55000]
23
24 print("\nEmployees with Salary > 55000:")
25 print(high_salary)
26
27 # Group by Department and calculate average salary
28 grouped = df.groupby("Department")["Salary"].mean()
29
30 print("\nAverage Salary by Department:")
31 print(grouped)
32
33 # Overall statistics
34 print("\nSummary Statistics:")
35 print(df.describe())
```

Screenshot 2

Program execution in Terminal/Command Prompt.

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\Alfido-project_2> pip install pandas
Collecting pandas
  Downloading pandas-3.0.3-cp313-cp313-win_amd64.whl.metadata (19 kB)
Collecting numpy>=1.26.0 (from pandas)
  Downloading numpy-2.5.0-cp313-cp313-win_amd64.whl.metadata (6.6 kB)
Collecting python-dateutil>=2.8.2 (from pandas)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting tzdata (from pandas)
  Downloading tzdata-2026.2-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading pandas-3.0.3-cp313-cp313-win_amd64.whl (9.8 MB)
 9.8/9.8 MB 3.2 MB/s 0:00:03
Downloading numpy-2.5.0-cp313-cp313-win_amd64.whl (12.4 MB)
 12.4/12.4 MB 7.4 MB/s 0:00:01
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading tzdata-2026.2-py2.py3-none-any.whl (349 kB)
Installing collected packages: tzdata, six, numpy, python-dateutil, pandas
Successfully installed numpy-2.5.0 pandas-3.0.3 python-dateutil-2.9.0.post0 six-1.17.0 tzdata-2026.2

[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
PS D:\Alfido-project_2> python data_analysis.py
Dataset:
   ID  Name Department  Salary  Age
0  101  Alice        HR  50000.0  25.0
1  102   Bob        IT  60000.0  28.0
2  103  Charlie       IT    NaN  30.0
3  104  David    Finance  55000.0  29.0
4  105   Eva        HR  52000.0   NaN

Missing Values:
ID          0
Name        0
Department  0
Salary      1
Age         1
dtype: int64

Cleaned Dataset:
   ID  Name Department  Salary  Age
0  101  Alice        HR  50000.000000  25.000000
1  102   Bob        IT  60000.000000  28.000000
2  103  Charlie       IT  56666.666667  30.000000
3  104  David    Finance  55000.000000  29.000000
4  105   Eva        HR  52000.000000  28.333333
5  106  Frank    Finance  58000.000000  31.000000
6  107  Grace        IT  65000.000000  27.000000

Employees with Salary > 55000:
   ID  Name Department  Salary  Age
1  102   Bob        IT  60000.000000  28.0
2  103  Charlie       IT  56666.666667  30.0
5  106  Frank    Finance  58000.000000  31.0
6  107  Grace        IT  65000.000000  27.0

Average Salary by Department:
Department
Finance    56500.000000
HR         51000.000000
IT         60555.555556
Name: Salary, dtype: float64

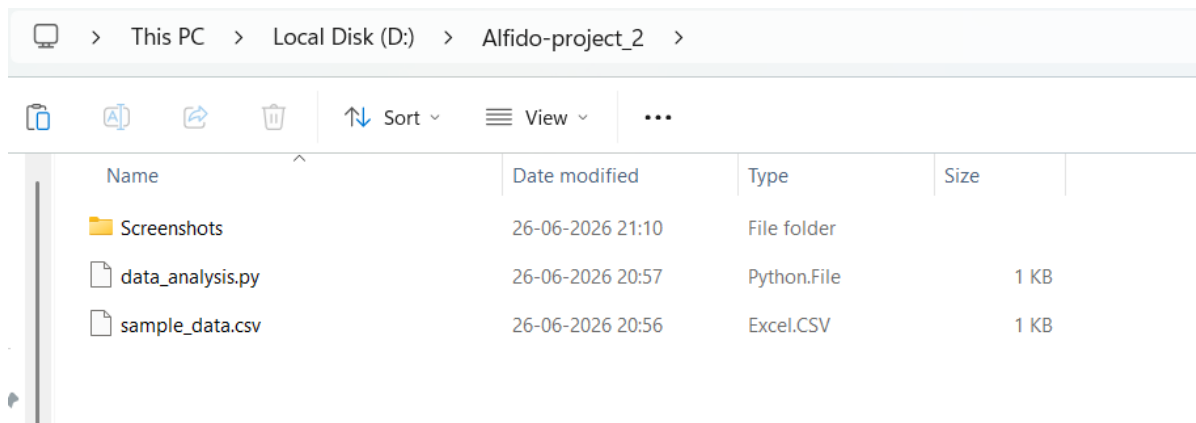
Summary Statistics:
      ID      Salary      Age
count  7.000000    7.000000    7.000000
mean   104.000000  56666.666667  28.333333
std     2.160247   5022.173058   1.972027
min    101.000000  50000.000000  25.000000
25%    102.500000  53500.000000  27.500000
50%    104.000000  56666.666667  28.333333
75%    105.500000  59000.000000  29.500000
max    107.000000  65000.000000  31.000000
PS D:\Alfido-project_2> █
```

Screenshot 3

Project folder after execution showing:

Project folder containing:

- data_analysis.py
- sample_data.csv
- screenshots folder



Data Cleaning

- Checked for missing values using isnull().
- Filled missing Salary values with the average salary.
- Filled missing Age values with the average age.

Data Analysis Performed

- Displayed the first five rows of the dataset.
- Filtered employees whose salary is greater than ₹55,000.
- Grouped data by Department.
- Calculated the average salary for each department.
- Displayed summary statistics of the dataset.

Insight Summary

- The dataset initially contained missing values in the Salary and Age columns.
- Missing values were replaced using the average values of the respective columns.
- Employees earning more than ₹55,000 were successfully filtered.
- The average salary was calculated for each department.
- The IT department had the highest average salary among all departments.

GitHub Repository

Repository Link:

<https://github.com/BhanuSri-19/Alfido-Tech-Task2>

Conclusion

This project demonstrates the use of the Pandas library for loading, cleaning, filtering, grouping, and analyzing data from a CSV file. It also provides meaningful insights from the dataset using simple and effective data analysis techniques.